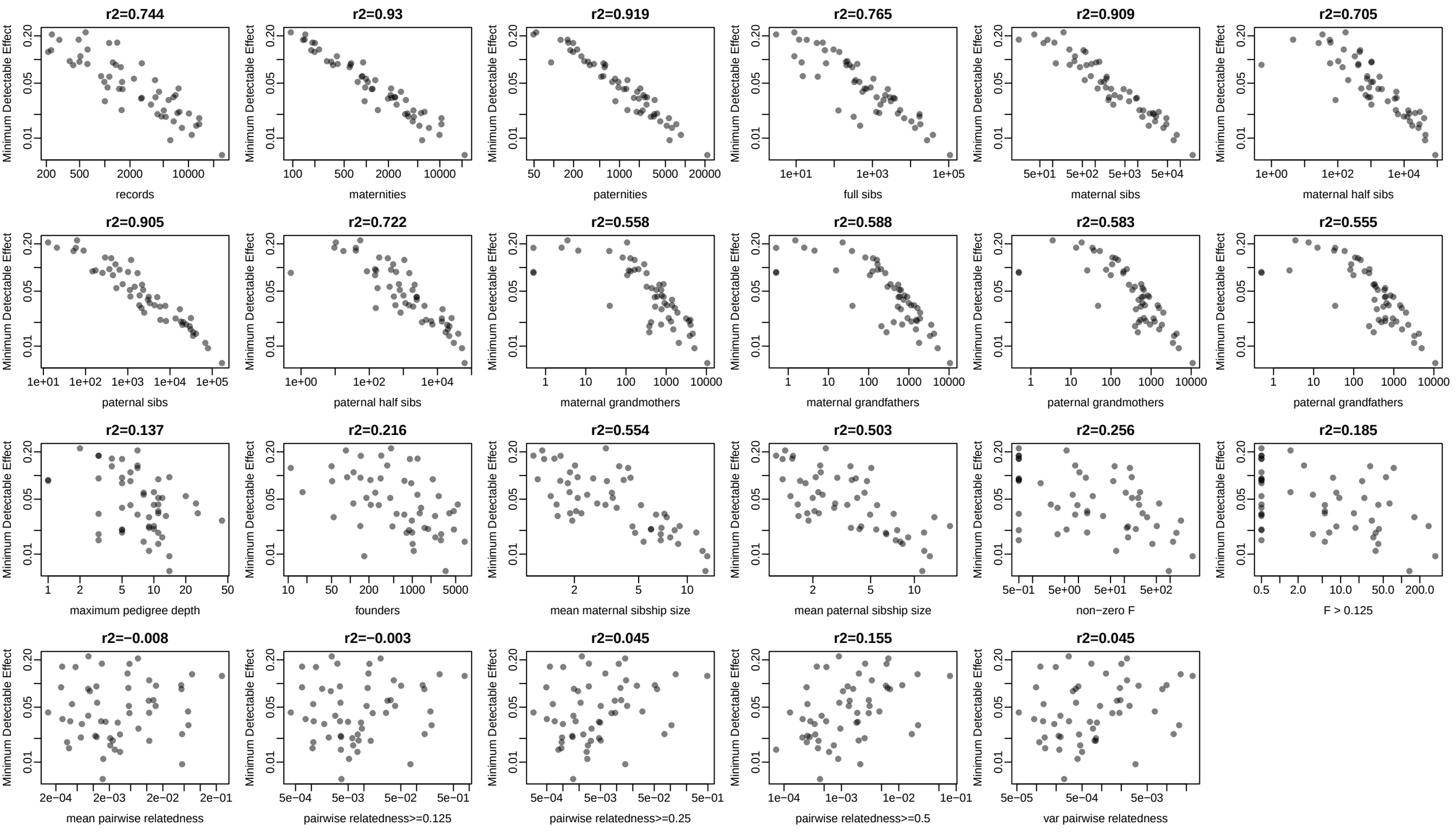

r2=0.744
Minimum Detectable Effect
0.20
0.05
0.01
200 500 2000 10000
records
r2=0.93
100 500 2000 10000
maternities
r2=0.919
50 200 1000 5000 20000
paternities
r2=0.765
1e+01 1e+03 1e+05
full sibs
r2=0.909
5e+01 5e+02 5e+03 5e+04
maternal sibs
r2=0.705
1e+00 1e+02 1e+04
maternal half sibs
r2=0.905
1e+01 1e+02 1e+03 1e+04 1e+05
paternal sibs
r2=0.722
1e+00 1e+02 1e+04
paternal half sibs
r2=0.558
1 10 100 1000 10000
maternal grandmothers
r2=0.588
1 10 100 1000 10000
maternal grandfathers
r2=0.583
1 10 100 1000 10000
paternal grandmothers
r2=0.555
1 10 100 1000 10000
paternal grandfathers
r2=0.137
1 2 5 10 20 50
maximum pedigree depth
r2=0.216
10 50 200 1000 5000
founders
r2=0.554
2 5 10
mean maternal sibship size
r2=0.503
2 5 10
mean paternal sibship size
r2=0.256
5e−01 5e+00 5e+01 5e+02
non−zero F
r2=0.185
0.5 2.0 10.0 50.0 200.0
F > 0.125
r2=−0.008
2e−04 2e−03 2e−02 2e−01
mean pairwise relatedness
r2=−0.003
5e−04 5e−03 5e−02 5e−01
pairwise relatedness>=0.125
r2=0.045
5e−04 5e−03 5e−02 5e−01
pairwise relatedness>=0.25
r2=0.155
1e−04 1e−03 1e−02 1e−01
pairwise relatedness>=0.5
r2=0.045
5e−05 5e−04 5e−03
var pairwise relatedness